

Xiaoqing PAN

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German Cancer Research Center
Im Neuenheimer Feld 280, Heidelberg

Re: Postdoctoral Position in Sex-Biased Cancer Development

Dear Prof. Dr. Odom,

I am writing to express my interest in the advertised postdoc position in your group. As a doctoral candidate in microbiology at the Hans-Knöll-Institute in Jena, Germany (thesis submitted 01.2026), my research mainly focused on double-stranded RNA (dsRNA) metabolism in infectious diseases. The project you posted focusing on comparative sex chromosomes and sex hormones contributing to cancer grabs my interest. Because the analysis of XX and XY genotype mice vs. XmO mice sounds interesting to me.

My PhD work was about host-fungal interactions, specializing in how human immune cells (PBMCs) responding to inflammatory triggers. With a focus on A-to-I RNA editing and RNA splicing analysis, I investigated dsRNA metabolism as it relates to the human immune response. We are currently working on a draft of this story, and, I successfully published a first-author review in *ACS Infectious Diseases*. I was also studying the RNA interference (RNAi) pathway in the human pathogen *Aspergillus fumigatus*. By investigating the sRNA landscape, we found that the fungus produced canonical sRNAs under the right condition. In this project, I generated and analyzed mRNA-, sRNA, tRNA-seq data and measured fungal biomass in the wet-lab to dissect how RNAi machinery processes dsRNA and regulates fungal development, work that resulted in first-author and co-author in *Molecular Microbiology* and *RNA*, respectively.

Based on the job description, the prerequisites align with my background in molecular biology. I have experience in analyzing genomic/transcriptomic data in an HPC environment, and I am interested to expand my skillset to include spatial and single-cell sequencing. While my expertise lies centers on the host-fungal system, the core principles of human immune pathways are broadly applicable. As a PhD student, I investigated the role of dsRNA in both fungal and host cells. Using multi RNA-seq methods, we discovered mechanisms for dsRNA that contributes to fungal growth and host immune response, respectively. I followed these studies with growth assays under different stress conditions, infection assays, molecular assays (e.g., PCR, DNA/RNA extraction, Northern/Southern Blot, Gibson cloning).

What draws me to your lab is the opportunity to dive into cancer research and learn comparative genomic analysis. I studied RNA regulation during my Masters and PhD, and I'd like to take the opportunity to complement my skillset DNA technologies together with your expertise. This is realistic since genomic analysis also relies on high-throughput sequencing technology. Second, I began to investigate host response in my PhD, which created my interest in better understanding human nucleic acid regulation. I believe this position is an opportunity to step forward to it. Third, I am seeking an international collaborative work environment, as I find integration into a vibrant team important for my development and performing high-quality science. DKFZ is the largest biomedical research institution in Germany. I also value the interdisciplinary nature of your team. I noticed your group includes members with diverse backgrounds (e.g., *Arabidopsis*, RNA chaperone proteins, planarians), and I believe that bringing my perspective from fungal genetics and host-pathogen immunology could contribute fresh insights to your projects. In the future, I want to improve my writing skills for a grant (e.g., Walter Benjamin Programme). My PhD supervisor, Dr. Matthew Blango (matthew.blango@leibniz-hki.de) and Dr. Olaf Kniemeyer (olaf.kniemeyer@leibniz-hki.de) have agreed to serve as referees. Thank you for your time and consideration.

Best,

Xiaoqing

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SUMMARY

Doctoral candidate in **Microbiology** specializing in RNA biology in the context of host-pathogen interactions. Proficient in R, Linux and NGS data processing within HPC environments (Slurm). Experienced in RNA-seq analysis, prediction of A-to-I RNA editing and alternative splicing, and microbial assays. Passionate about integrating bioinformatics to address questions in biotechnology and immunology. Published five papers and two in preparation.

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany 2026

Supervisor: [Dr. Matthew Blango](#) (RNA Biology of Fungal Infections)

Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs (submitted)

MSc | Bioengineering, Chinese Academy of Sciences, China 2021

Supervisor: Dr. Kuaifei Xia (Agriculture & Biotechnology Research Center)

Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively

PUBLICATIONS

- **Pan X***, Kelani AA*, Schrettenbrunner L, Prabakar S, Israni B, Blango MG. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microbiol 1-12.
- **Pan X***, Bruch A*, Blango MG. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis 10(12), 4017-4029.
- Xia K*, **Pan X***, Chen H, Xu X, Zhang M. (2023). XOO-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of *OsARAB* in rice. Phytopathol Res 5, 22.
- Xia K*, **Pan X***, Zeng X, Zhang M. (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting *OsOPF3*, *OsNPF2.4*, and *OsAGO1a*, respectively. J. Plant Physiol 208, 153905.
- Kelani AA, Bruch A, Rivieccio F, Visser C, Krüger T, Weaver D, **Pan X**, Schaeuble S, Panagiotou G, Knimeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. RNA 29(7), 1033-1050.

PROFESSIONAL EXPERIENCES

Human PBMC transcriptome changes upon fungal stimulations

- Designing a high-throughput **Snakemake workflow** to quantify differential expression and alternative splicing events, characterizing the dynamic response of human PBMCs to fungal pathogens.
- Leading a collaboration with Fritz-Lipmann-Institut, Germany
- DOI: [10.1021/acsinfecdis.4c00598](https://doi.org/10.1021/acsinfecdis.4c00598)

RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Develop both sRNA- and mRNA-seq pipeline for differential expression gene analysis.
- Experimental validation support for determining **the role of RNAi machinery and sRNA** using Northern Blot, PCR, qPCR assays.
- Collaborate with The University of Manchester, the University of Jena
- DOI: <https://doi.org/10.1111/mmi.70042> and DOI: [10.1261/rna.079350.122](https://doi.org/10.1261/rna.079350.122)

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analysis of human extracellular small RNA transcriptome.
- Collaborate with the University of Jena.

Non-coding RNA analysis of rice-Xoo interaction

- Experimental validation support for determining **non-coding RNAs (sRNAs, lncRNAs, circRNAs) in rice responding to fungal infections** using PCR, qPCR, LC-MS, assays.
- Collaborate with Sun Yat-Sen University, China

- DOI: [10.1186/s42483-023-00188-8](https://doi.org/10.1186/s42483-023-00188-8) and DOI: [10.1016/j.jplph.2022.153905](https://doi.org/10.1016/j.jplph.2022.153905)

Teaching

- Microbiology and Molecular Biology - MMB005, 2022 - 2025
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TECHNICAL SKILLS

- **Bioinformatics:** HPC, Snakemake, Genomics, RNA-seq, SHAPE-seq, Alternative splicing, Data visualization
 - **Molecular Biology:** DNA/RNA/protein isolation, Genetic Engineering, Northern Blot, qPCR, PCR, LC-MS, Gibson Assembly Cloning, Cultivation of microorganisms
 - **Programming & Tools:** R, python, Bash, Git, Docker, Linux
 - **Language:** English (fluent), German (B1), Chinese (native)
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CONFERENCES

- **2025 - Oral presentation:** German speaking Mycological Society (DMykG), Germany
 - **2025 - Poster:** Gordon Research Seminar - RNA Editing, Italy
 - **2024 - Oral presentation:** Jena RNA Club Symposium, Germany
 - **2024 - Poster:** RNA Society Meeting, UK **2023 - Poster:** EMBL - The Non-coding Genome Symposium, Germany
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AWARDS & SCHOLARSHIP

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|---|-------------|
| • DAAD Scholarship | 2021 - 2025 |
| • Jena School for Microbial Communication (JSMC) travel grant | 2023 |
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REFEREES

- Dr. Matthew Blango (matthew.blango@leibniz-hki.de)
- Dr. Olaf Kniemeyer (olaf.kniemeyer@leibniz-hki.de)